The area of a parallelogram and of a triangle

Cut a corner off and slide it across — and the parallelogram becomes a rectangle.

LESSON 47–48 | 2 × 40 MIN | GEOMETRY 8, ТЕМЫ 47–48 | STAGE 9 · 7.1–7.2

LESSON CLOCK

6' | 14' | 12' | 4' | 4'

Cut and slide a paper parallelogram	6 min
The parallelogram formula	14 min
The triangle formula	12 min
Two sides and an angle	4 min
Homework	4 min

LEARNING OBJECTIVES

- Derive and use $S = a \cdot h$ for a parallelogram.
- Derive and use $S = ½\, a \cdot h$ for a triangle.
- Use $S = ½\, ab \sin C$ when two sides and the included angle are known.
- Recognise that triangles on the same base between the same parallels have equal areas.

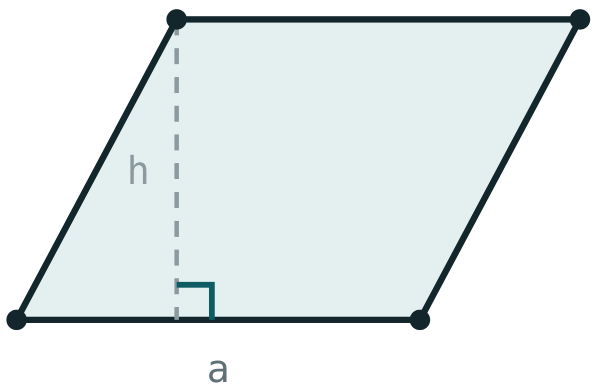
TEXTBOOK

National	Темы 47–48, pp. 106–113
Cambridge	Learner’s Book pp. 142–155
Workbook	Workbook 7.1–7.2

01 Explanation

The parallelogram

Drop a perpendicular from one vertex, cut off the right triangle it makes and slide it to the other end. Nothing is lost and nothing is added — the parallelogram has become a rectangle with the same base and the same height.



Cut the triangle off one end and slide it to the other: the parallelogram becomes a rectangle of the same base and height.

AREA OF A PARALLELOGRAM

$$S = a \cdot h_a$$

where h_a is the height **to the side *a***.

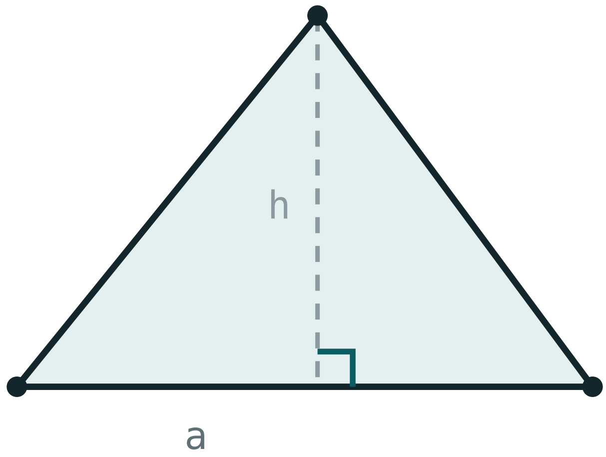
HEIGHT, NOT SLANT SIDE

In a parallelogram with sides 8 and 5 the area is **not** 40 unless the angle is right. The height is the perpendicular distance between the two parallel sides, and it is always shorter than the slant side.

Since $S = a \cdot h\_a = b \cdot h\_b$, a longer side always has a shorter height. That single equation solves a whole family of problems.

The triangle

Any triangle is exactly half of a parallelogram: copy it, turn the copy through $180°$ and join the two along a side.



Two copies of a triangle make a parallelogram, so a triangle is half of base times height.

$$S = \frac{1}{2}\, a \cdot h_a$$

For a right-angled triangle the two legs serve as base and height, so $S = \frac{1}{2}ab$ immediately.

TWO SIDES AND THE ANGLE BETWEEN THEM

$$S = \frac{1}{2}\, ab \cdot \sin C$$

because the height to a is $b \sin C$. With $C = 90°$ this is the right-angle formula again, since $\sin 90° = 1$.

Same base, same parallels

The height of a triangle does not depend on where the apex sits along a line parallel to the base. So:

EQUAL AREAS

Triangles with the **same base** and apexes on the **same line parallel** to it have **equal areas** — however different they look.

Two consequences worth remembering:

- A **median** divides a triangle into two of equal area — the two halves share a height and have equal bases.
- The diagonals of a parallelogram cut it into four triangles of equal area.

02 Worked examples

EXAMPLE 1

A parallelogram has sides *10 cm* and *6 cm*, and the height to the *10 cm* side is *4 cm*. Find its area and the other height.

1. $S = 10 \cdot 4 = 40\ cm^2$

   Base times its own height.

2. $40 = 6 \cdot h_b$

   The same area, measured from the other side.

3. $h_b = \frac{40}{6} \approx 6.67\ cm$

   Shorter side, taller height.

ANSWER

$40\ cm^2$, $h_b \approx 6.67\ cm$

EXAMPLE 2 **Find the area of a triangle with sides *7 cm* and *8 cm* and an included angle of *30°*.**

1. $S = \frac{1}{2} \cdot 7 \cdot 8 \cdot \sin 30°$
2. $\sin 30° = 0.5$
3. $S = \frac{1}{2} \cdot 56 \cdot 0.5$
4. $= 14\ cm^2$

ANSWER

$14\ cm^2$

EXAMPLE 3 **A triangle has area *54 cm²* and base *12 cm*. Find its height, and the area of each half made by the median to that base.**

1. $54 = \frac{1}{2} \cdot 12 \cdot h$
2. $54 = 6h,\ h = 9\ cm$
3. $54 \div 2 = 27\ cm^2$

   A median always halves the area.

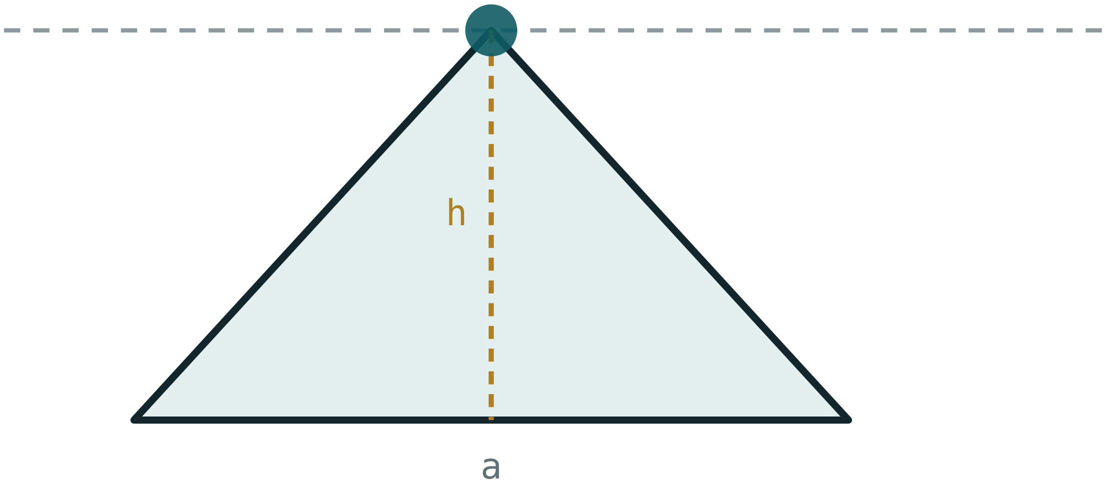
ANSWER

$h = 9\ cm$; each half $27\ cm^2$

03 Interactive model

Slide the apex along the top line: the shape changes completely, the area does not.

Same base, same height, same area



base a **220** | height h **120** | area ½ah **13200** | perimeter **545.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Parallelogram	Parallelogramm	Параллелограмм
Base	Asos	Основание
Height (altitude)	Balandlik	Высота
Corresponding height	Mos balandlik	Соответствующая высота
Triangle	Uchburchak	Треугольник
Included angle	Orasidagi burchak	Угол между сторонами
Equal areas	Teng yuzalar	Равновеликие
Median	Mediana	Медиана
Between the same parallels	Bir xil parallellar orasida	Между теми же параллельными

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A parallelogram with base 9 and height 4 has area:

- 13
- 18
- 36
- 72

A triangle with base 10 and height 6 has area:

- 60
- 30
- 16
- 15

A parallelogram has sides 8 and 5. Its area is:

- 40
- 26
- less than 40
- 20

A median divides a triangle into two parts of:

- unequal area
- equal area
- equal perimeter
- equal angles

For sides 6 and 10 with an included angle of $30°$, the area is:

- 30
- 15
- 60
- 7.5

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *Find the area of a parallelogram with base 7 and height 5.*

   ANSWER *35*

2. *Find the area of a triangle with base 8 and height 3.*

   ANSWER *12*

3. *Find the area of a right triangle with legs 6 and 9.*

   ANSWER *27*

4. *A triangle has area 20 and base 10. Find its height.*

   ANSWER *4*

5. *A parallelogram has area 48 and base 12. Find its height.*

   ANSWER *4*

6. *Find the area of a triangle with base 5 and height 5.*

   ANSWER *12.5*

7. *A median divides a triangle of area 36. Find each part.*

   ANSWER *18*

Medium · the standard question

7 PROBLEMS

1. *A parallelogram has sides 12 and 9, and the height to the 12-side is 6. Find the other height.*

   ANSWER *8*

2. *Find the area of a triangle with sides 10 and 12 and included angle 30°.*

   ANSWER *30*

3. *A triangle has area 42 cm² and height 7 cm. Find its base.*

   ANSWER *12 cm*

4. *The diagonals of a parallelogram of area 60 cut it into four triangles. Find each area.*

   ANSWER *15*

5. *Find the area of a triangle with vertices (0;0), (6;0), (0;8).*

   ANSWER *24*

6. *A parallelogram has base 15 cm and area 90 cm². Find its height.*

   ANSWER *6 cm*

7. *Find the area of a triangle with sides 8 and 8 and included angle 90°.*

   ANSWER *32*

Hard · stretch and reason

7 PROBLEMS

1. *A parallelogram has sides 10 and 8 with an angle of 30°. Find its area.*

   ANSWER $10 \cdot 8 \cdot \sin 30° = 40$

2. *A triangle has sides 13, 14, 15. Find its area.*

   ANSWER *84* — the height to the 14-side is 12

3. *The area of a triangle is 24 cm² and two of its sides are 8 cm and 12 cm. Find the angle between them.*

   ANSWER $\sin C = 0.5$, so $C = 30°$ (or $150°$)

4. *ABCD is a parallelogram and P is any point on AB. Show that the area of triangle PCD is half the area of ABCD.*

   ANSWER The triangle has base CD and its apex lies on the parallel line AB, so its height is the height of the parallelogram; its area is $\frac{1}{2} \cdot CD \cdot h$.

5. *Find the area of a triangle with vertices (1;1), (5;2), (3;7).*

   ANSWER *11* — box $4 \cdot 6 = 24$ less three corner triangles

6. *A parallelogram has diagonals of 10 and 6 crossing at 90°. Find its area.*

   ANSWER *30* — it is a rhombus, area $\frac{1}{2} d_1 d_2$

7. *Two triangles have the same base and equal areas. Must their apexes lie on a line parallel to that base?*

   ANSWER Yes — equal area with equal base forces equal height, so both apexes are the same distance from the base line, on one of the two parallels.

07 Homework

Homework — 6 problems

Geometry 8, Темы 47–48, pp. 106–113. Mark the height on every diagram.

1. *Find the area of a parallelogram with base 14 and height 9.*

2. *Find the area of a triangle with base 16 and height 7.5.*
3. *A parallelogram has sides 15 and 10; the height to the 15-side is 8. Find the other height.*
4. *Find the area of a triangle with sides 9 and 14 and included angle 30°.*
5. *A triangle has area 63 cm² and base 18 cm. Find its height.*
6. *Find the area of the triangle with vertices (0;0), (10;0), (4;6).*